

Yechan Yang

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Interest

Interface, AI, Biocomputing

Education

- Seoul National University**, PhD in Biomedical Engineering 2026.03 – Present
- Advisor: Prof. Seongjun Park
- Korea University**, MS in Computer Science and Engineering 2022.09 – 2024.08
- Advisor: Prof. Gerard Jounghyun Kim
 - GPA: 4.44/4.50 (99.4)
- Korea University**, BS in Mechanical Engineering 2016.03 – 2022.08
- 2 years absence to fulfill military service (2019–2021)
 - GPA: 3.82/4.50 (93.2)

Research Experience

- Seoul National University** 2025.09 – Present
- Research Assistant (Advisor: Seongjun Park)
 - Research Focus: Brain-Computer Interface, Biocomputing
- Korea Institute of Science and Technology** 2024.08 – 2025.08
- Research Assistant (Advisor: Byounghyun Yoo)
 - Research Focus: Signal Processing
- Korea University** 2022.08 – 2024.08
- Graduate Research Assistant (Advisor: Gerard Jounghyun Kim)
 - Research Focus: VR, Cybersickness, Human-Computer Interaction

Publications (*co-first authorship indicated)

Papers

- **Y. Yang***, E. Chang*, and B. Yoo. *Trunk Motion Reflects Center-of-Pressure Dynamics and Supports Markerless Single-Camera Postural Sway Assessment*. Scientific Reports (IF = 4.9, Q1).
- Y. Lee*, S. Jeon*, M. Kim*, J. Won, J. Yeo, K. Jo, S. Lee, K. Park, **Y. Yang**, C. Kim, G.-H. Lee, S. Ryu, S. Jo, and S. Park. *1-D Collocated Dual-Gradient Sensory Fibers for Comprehensive Sensing and Decoding of Complex Human Motion and Physiology*. Advanced Materials (IF = 29.1, Q1).
- M. Kim, J. Ryu*, **Y. Yang***, and G. Kim. *JourneyVR: Designing for Continuous Workflow Experience in Virtual Reality*. The ACM Conference on Human Factors in Computing Systems (CHI 2026).
- **Y. Yang**, J. Park, G. Kim, and H. Kim. *Effects of Mini-Map Usage and Spatial Ability on Cybersickness in Virtual Reality Navigation*. International Journal of Human-Computer Interaction (IF = 4.9, Q1).
- S. Park*, **Y. Yang***, M. Kim, and B. Yoo. *StreamSplat: A Hybrid Client-Server Architecture for Neural Graphics using Depth-based Fusion on the Web*. The International Conference on 3D Web Technology (Web3D 2025).
- J. Park, **Y. Yang**, G. Kim, and H. Kim. *The Impact of Observer Presence on Trainees' Mental States and Performance in Remote Military Training with Virtual Humans in Mixed Reality Environment*. The ACM Conference on Human Factors in Computing Systems (CHI 2025).
- **Y. Yang***, S. Kang*, M. Kim, G. Kim, and H. Kim. *BalanceVR: VR-Based Balance Training to Increase Tolerance to VR Sickness*. Virtual Reality (IF = 5.0, Q1).
- J. Lee, H. Kim, **Y. Yang**, and G. Kim. *RPG: Rotation Technique in VR Locomotion Using Peripheral Gaze*. The ACM Symposium on Eye Tracking Research & Applications (ETRA 2024).

Posters

- M. Kim, **Y. Yang**, and G. Kim. *JourneyVR: Experience Design for Continuous Timed Activities in Virtual Reality*. IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025).
- **Y. Yang**, H. Kim, and G. Kim. *u-DFOV: User-Activated Dynamic Field of View Restriction for Managing Cybersickness and Task Performance*. The IEEE Conference on Virtual Reality and 3D User Interfaces (IEEEVR 2024).
- **Y. Yang**, H. Kim, J. Kim, and G. Kim. *PianoFMS: Real-time Evaluation of Cybersickness by Keyboard Fingering*. The IEEE Conference on Virtual Reality and 3D User Interfaces (IEEEVR 2024).
- **Y. Yang**, H. Kim, and G. Kim. *The Effects of Customized Strategies for Reducing VR Sickness*. The ACM Symposium on Virtual Reality Software and Technology (VRST 2023).
- S. Song, **Y. Yang**, J. Lee, and G. Kim. *FakeBand: Virtual Band Music Performance with Balanced Interface for Individual Score/Rhythm Play and Inter-player Expression Coordination*. The IEEE Conference on Virtual Reality and 3D User Interfaces (IEEEVR 2023).

Teaching Experience

Teaching Assistant, Korea University, Seoul, Korea

- Data Science Spring 2024
- Creative Software Startup Methodology Fall 2023

Honors and Scholarships

Doctoral Excellence Scholarship (STEM), Korea Student Aid Foundation 2026.09–2030.09

Semester High Honors, Korea University Spring 2018 & Fall 2021

Awards

Minister of Health and Welfare Award (1st Place), 7th NRC Assistive Device Hackathon 2026.08

SNU Grand Quest Maker, Seoul National University 2026.06

Best Paper Award, Web3D 2025 2025.09

Outstanding Research Award, Korea Electronics Association (KEA) 2023.08

Outstanding Exhibition Center Award, ITRC 2023.04

IITP President's Award (4th Place) ICT Challenge 2022 2022.09

Technical Skills

Artificial Intelligence: Machine Learning, Time-Series Analysis, Signal Processing

Programming: Python, MATLAB, C/C++, C#, Unity

Neural Engineering: Stereotaxic Surgery, Neural Cell Culture, Neural Recording

XR & Prototyping: AR/VR/XR, Arduino, CAD, 3D Printing